



UNIVERSITY OF GENEVA

Faculty of Science

Exercise #1

Introduction to Computational Finance
14X030

Michel Jean Joseph Donnet



1 Computing returns

The following series describes the value of a portfolio at the end of each month:

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	t
0	1	2	3	4	5	6	7	8	9	10	11	12	$P(t)$
100	101	102	103	104	105	106	107	108	109	110	111	112	

The initial value of the portfolio being 100, the annual return is:

$$R_a = R_{12}(12) = \frac{P(12) - P(0)}{P(0)} = 12\%$$

1. Compute monthly returns $\{R(t) \mid t = 1, \dots, 12\}$.

The monthly returns are the following:

$$\begin{aligned} R_1 &= \frac{P(1)-P(0)}{P(0)} = \frac{101-100}{100} = \frac{1}{100} \\ R_2 &= \frac{P(2)-P(1)}{P(1)} = \frac{102-101}{101} = \frac{1}{101} \\ R_3 &= \frac{P(3)-P(2)}{P(2)} = \frac{103-102}{102} = \frac{1}{102} \\ R_4 &= \frac{P(4)-P(3)}{P(3)} = \frac{104-103}{103} = \frac{1}{103} \\ R_5 &= \frac{P(5)-P(4)}{P(4)} = \frac{105-104}{104} = \frac{1}{104} \\ R_6 &= \frac{P(6)-P(5)}{P(5)} = \frac{106-105}{105} = \frac{1}{105} \\ R_7 &= \frac{P(7)-P(6)}{P(6)} = \frac{107-106}{106} = \frac{1}{106} \\ R_8 &= \frac{P(8)-P(7)}{P(7)} = \frac{108-107}{107} = \frac{1}{107} \\ R_9 &= \frac{P(9)-P(8)}{P(8)} = \frac{109-108}{108} = \frac{1}{108} \\ R_{10} &= \frac{P(10)-P(9)}{P(9)} = \frac{110-109}{109} = \frac{1}{109} \\ R_{11} &= \frac{P(11)-P(10)}{P(10)} = \frac{111-110}{110} = \frac{1}{110} \\ R_{12} &= \frac{P(12)-P(11)}{P(11)} = \frac{112-111}{111} = \frac{1}{111} \end{aligned}$$

2. Compute the annual return R_a from the monthly returns (you should find 12% again). Compare it to the sum of the monthly returns.

$$1 + R_a = \prod_{i=1}^{12} (1 + R_i) \Leftrightarrow R_a = \left[\prod_{i=1}^{12} (1 + R_i) \right] - 1 = 12\%$$

```
R = [1/(100 + i) for i in range(12)]
R_a = 1
for R_i in R:
    R_a *= (1 + R_i)
R_a -= 1
print(f"R_a = {R_a:.2f}")
```

$R_a = 0.12$

3. Compute the average monthly return R_m and compare it to the average of the monthly returns.

- Average monthly return: $R_m = \left(1 + \frac{R_a}{12}\right)^{12} - 1 = 2\%$



```
R_m = (1 + 12/100 * 1/12)**2
R_m -= 1
print(f"R_m = {R_m:.2f}")
```

`R_m = 0.02`

- Average of the monthly returns: $\frac{1}{12} \sum_{i=1}^{12} R_i = \frac{1}{12} \left(\frac{1}{100} + \frac{1}{101} + \dots + \frac{1}{112} \right) = 1\%$

```
R = [1/(100 + i) for i in range(12)]
average = 0
for R_i in R:
    average += R_i
average /= 12
print(f"Average of monthly returns: {average:.2f}")
```

Average of monthly returns: `0.01`

So it means that the average of the monthly returns is different from the average monthly return.

2 Continuously compounded return

Consider a constant annual interest rate of $R = 2.11\%$ with a continuous compounding.

1. With an initial amount of \$10 000, what will be the value in:
 - 1 year?
 - 5 years?
 - 10 years?

We note $n = 1$ year, FV_n the Future Value at year n and $PV = 10000$ the Present Value.

The continuous compounding formula is $FV_n = PV \cdot e^{Rn}$

So we have:

- $FV_1 = PV \cdot e^R = 10000 \cdot e^{0.0211} = 10213.24\$$
- $FV_5 = PV \cdot e^{R \cdot 5} = 10000 \cdot e^{0.0211 \cdot 5} = 11112.66\$$
- $FV_{10} = PV \cdot e^{R \cdot 10} = 10000 \cdot e^{0.0211 \cdot 10} = 12349.12\$$

2. What is the initial amount to have a final amount of \$10 000 in:
 - 1 year?
 - 3 years?
 - 10 years?

Now, we know the final amount $FV_n = 10000\$$. So the initial amount PV is:

$$FV_n = PV \cdot e^{Rn} \Leftrightarrow PV = FV_n e^{-Rn}$$

So to have a final amount of 10000 after n years, the initial amount must be:

- For $n = 1$: $PV = FV_1 e^{-R} = 10000 e^{-0.0211} = 9791.21\$$
- For $n = 3$: $PV = FV_3 e^{-R \cdot 3} = 10000 e^{-0.0211 \cdot 3} = 9386.62\$$
- For $n = 10$: $PV = FV_{10} e^{-R \cdot 10} = 10000 e^{-0.0211 \cdot 10} = 8097.74\$$



3 Portfolio of Microsoft and Starbucks stock

You purchase 10 shares of each Microsoft and Starbucks stock at the end of month $t - 1$ at prices $P_{\text{msft}(t-1)} = \$85$ and $P_{\text{sbux}(t-1)} = \$30$.

1. Compute $V(t - 1)$, the initial value of the portfolio.

The initial value $V(t - 1)$ of the portfolio is:

$$V(t - 1) = 10 \cdot P_{\text{msft}(t-1)} + 10 \cdot P_{\text{sbux}(t-1)} = 10 \cdot 85 + 10 \cdot 30 = 850 + 300 = 1150\$$$

2. Compute the portfolio shares α_{msft} and α_{sbux} .

- The portfolio share α_{msft} has a value of: $\alpha_{\text{msft}} = \frac{10}{10+10} = \frac{1}{2}$
- The portfolio share α_{sbux} has a value of: $\alpha_{\text{sbux}} = \frac{10}{10+10} = \frac{1}{2}$

Consider now that at the end of month t , the prices are $P_{\text{msft}(t)} = \$90$ and $P_{\text{sbux}(t)} = \$28$.

3. Compute $R_{\text{msft}(t)}$ and $R_{\text{sbux}(t)}$, the one-period return of Microsoft and Starbucks stocks.

- $R_{\text{msft}(t)} = \frac{P_{\text{msft}(t)} - P_{\text{msft}(t-1)}}{P_{\text{msft}(t-1)}} = \frac{90-85}{85} \approx 5.9\%$
- $R_{\text{sbux}(t)} = \frac{P_{\text{sbux}(t)} - P_{\text{sbux}(t-1)}}{P_{\text{sbux}(t-1)}} = \frac{28-30}{30} \approx -6.7\%$

4. Compute the one-period return of the portfolio $R(t)$ and its value $V(t)$ at the end of month t .

- The one-period return of the portfolio $R(t)$ is:

$$R(t) = \sum_{i \in \{\text{msft}, \text{sbux}\}} \alpha_i R_i = \alpha_{\text{msft}} R_{\text{msft}} + \alpha_{\text{sbux}} R_{\text{sbux}} = \frac{1}{2} \cdot 5.9\% - \frac{1}{2} \cdot 6.7\% = -0.4\%$$

- The value $V(t)$ of the portfolio is:

$$V(t) = V(t - 1)(1 + R(t)) = 1150(1 - 0.004) = 1145.4\$$$

4 Present / Future value

Consider a constant annual return $R = 4.5\%$, semi-annually compounded (i.e. two payments a year).

1. With an initial amount of \$10 000, what will be the value in:

- 1 year?
- 5 years?
- 10 years?

We note $n = 1$ year, FV_n the Future Value at year n and $PV = 10000$ the Present Value.

For a semi-annual compounded, we have:



$$FV_n = PV \cdot \left(1 + \frac{R}{2}\right)^{2 \cdot n}$$

So we have:

- $FV_1 = PV \cdot \left(1 + \frac{R}{2}\right)^2 = 10000 \cdot \left(1 + \frac{0.045}{2}\right)^2 = 10455.06\$$
- $FV_5 = PV \cdot \left(1 + \frac{R}{2}\right)^{2 \cdot 5} = 10000 \cdot \left(1 + \frac{0.045}{2}\right)^{10} = 12492.03\$$
- $FV_{10} = PV \cdot \left(1 + \frac{R}{2}\right)^{2 \cdot 10} = 10000 \cdot \left(1 + \frac{0.045}{2}\right)^{20} = 15605.09\$$

2. What is the initial amount to have a final amount of \$10 000 in:

- 1 year?
- 3 years?
- 10 years?

We know the final amount $FV_n = 10000\$$

We want to find PV .

For a semi-annual compounded, we have:

$$FV_n = PV \cdot \left(1 + \frac{R}{2}\right)^{2 \cdot n} \Leftrightarrow PV = FV_n \cdot \left(1 + \frac{R}{2}\right)^{-2 \cdot n}$$

So to have a final amount of 10000 after n years, the initial amount must be:

- For $n = 1$: $PV = FV_1 \cdot \left(1 + \frac{R}{2}\right)^{-2} = 10000 \cdot \left(1 + \frac{0.045}{2}\right)^{-2} = 9564.74\$$
- For $n = 3$: $PV = FV_3 \cdot \left(1 + \frac{R}{2}\right)^{-2 \cdot 3} = 10000 \cdot \left(1 + \frac{0.045}{2}\right)^{-6} = 8750.24\$$
- For $n = 10$: $PV = FV_{10} \cdot \left(1 + \frac{R}{2}\right)^{-2 \cdot 10} = 10000 \cdot \left(1 + \frac{0.045}{2}\right)^{-20} = 6408.16\$$

3. **Optional:** Let R be the one-period return, n the number of periods, and m the number of payments per period. How would you prove that:

$$\lim_{m \rightarrow +\infty} \left(1 + \frac{R}{m}\right)^{mn} = e^{Rn}$$